# SECTION 1: DOUBLE RIVETED LAP JOINT UNDER TENSION, SHEAR, AND CRUSHING STRESSES

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**System Parameters:** Plate Thickness $t = 15$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 75$ mm, Ultimate Tensile Stress $\sigma_\mathrm{u} = 400$ MPa, Ultimate Shear Stress $\tau_\mathrm{u} = 320$ MPa, Ultimate Crushing Stress $\sigma_\mathrm{cu} = 640$ MPa, Factor of Safety FOS $= 4$.

**Design Protocol:** Evaluate plate tearing, rivet shearing ($n = 2$), and rivet crushing strengths to determine minimum rupturing force $P_\mathrm{u}$ per pitch; compute working load $P = P_\mathrm{u}$ / FOS and evaluate working stresses.

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**1. Tearing Strength of Plate ($P_t$)**

Plate tearing capacity per pitch length

$$P_t = (p - d)t \cdot \sigma_\mathrm{u}$$
$$= (75 - 25) \times 15 \times 400$$
$$= 300000 \text{ N}$$
$$\Rightarrow P_t = 300 \text{ kN}$$

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**2. Shearing Strength of Rivets ($P_s$)**

Single shear capacity of 2 rivets per pitch length

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau_\mathrm{u}$$
$$= 2 \times \frac{\pi}{4}(25)^2 \times 320$$
$$= 314160 \text{ N}$$
$$\Rightarrow P_s = 314.16 \text{ kN}$$

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**3. Crushing Strength of Rivets ($P_c$)**

Bearing capacity of 2 rivets per pitch length

$$P_c = n \cdot d \cdot t \cdot \sigma_{\text{cu}}$$
$$= 2 \times 25 \times 15 \times 640$$
$$= 480000 \text{ N}$$
$$\Rightarrow P_c = 480 \text{ kN}$$

**4. Rupturing Force per Pitch ($P_u$)**

Minimum strength governing joint rupture

$$P_u = \min(P_t, P_s, P_c)$$
$$= \min(300, 314.16, 480)$$
$$\Rightarrow P_u = 300 \text{ kN}$$

**5. Safe Working Load ($P$)**

Applied force per pitch at FOS = 4

$$P = \frac{P_u}{\text{FOS}}$$
$$= \frac{300}{4}$$
$$= 75 \text{ kN}$$
$$= 75000 \text{ N}$$

**6. Actual Tensile Stress ($\sigma_t$)**

Induced tensile stress on main plate

$$\sigma_t = \frac{P}{(p-d)t}$$

$$= \frac{75000}{(75-25)\times 15}$$

$$= 100 \text{ MPa}$$

**7. Actual Shear Stress ($\tau$)**

Induced shear stress in rivets

$$\tau = \frac{P}{n \cdot \frac{\pi}{4} d^2}$$

$$= \frac{75000}{2 \times \frac{\pi}{4}(25)^2}$$

$$= 76.4 \text{ MPa}$$

**8. Actual Crushing Stress ($\sigma_c$)**

Induced crushing stress on rivets

$$\sigma_c = \frac{P}{n \cdot d \cdot t}$$

$$= \frac{75000}{2 \times 25 \times 15}$$

$$= 100 \text{ MPa}$$

**Design Output**

**Rupturing Force Pu = 300 kN | Working Load P = 75 kN | Working Stresses: sigma_t = 100 MPa, tau = 76.4 MPa, sigma_c = 100 MPa**

# SECTION 1: DOUBLE RIVETED LAP JOINT — MECHANICAL DIAGRAM

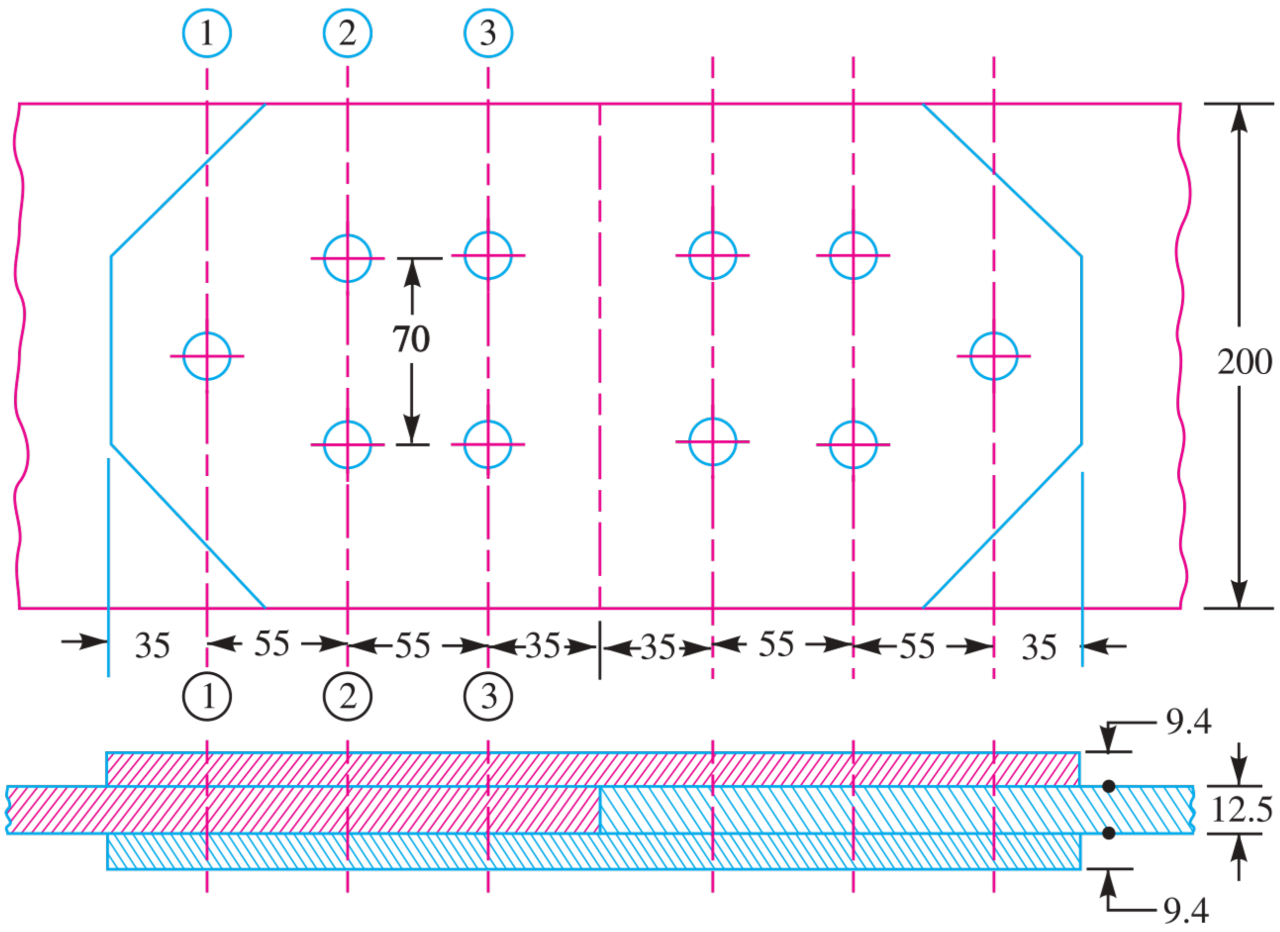


**Fig. 9.20.** All dimensions in mm.

**Fig. 9.20: Double Riveted Lap Joint Scheme**

# SECTION 2: EFFICIENCY OF SINGLE AND DOUBLE RIVETED LAP JOINTS

**System Parameters:** Plate Thickness $t = 6\ \mathrm{mm}$, Rivet Diameter $d = 20\ \mathrm{mm}$, Case 1 Pitch $p_1 = 50\ \mathrm{mm}$ ($n = 1$), Case 2 Pitch $p_2 = 65\ \mathrm{mm}$ ($n = 2$), Allowable Stresses: Tensile $\sigma_t = 120\ \mathrm{MPa}$, Shear $\tau = 90\ \mathrm{MPa}$, Crushing $\sigma_c = 180\ \mathrm{MPa}$.

**Design Protocol:** Calculate un-punched solid plate strength, tearing, shearing, and crushing strengths for single and double riveted lap joints; compute joint efficiencies.

**1. Solid Plate Strength — Single Lap ($P$)**

Strength of un-punched plate ($p = 50\ \mathrm{mm}$)

$$P = p \cdot t \cdot \sigma_t$$
$$= 50 \times 6 \times 120$$
$$= 36000\ \mathrm{N}$$

**2. Tearing Strength — Single Lap ($P_t$)**

Net section tensile capacity

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (50 - 20) \times 6 \times 120$$
$$= 21600\ \mathrm{N}$$

**3. Shearing Strength — Single Lap ($P_s$)**

Single shear capacity ($n = 1$)

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1 \times \frac{\pi}{4}(20)^2 \times 90$$
$$= 28274\ \mathrm{N}$$

**4. Crushing Strength — Single Lap ($P_c$)**

Bearing capacity ($n = 1$)

$$P_c = n \cdot d \cdot t \cdot \sigma_c$$
$$= 1 \times 20 \times 6 \times 180$$
$$= 21600 \text{ N}$$

**5. Efficiency — Single Riveted Lap Joint ($\eta_1$)**

Ratio of minimum strength to solid strength

$$\eta_1 = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$
$$= \frac{21600}{36000} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta_1} = \mathbf{60.0}\%$$

**6. Solid Plate Strength — Double Lap ($P$)**

Strength of un-punched plate ($p = 65 \text{ mm}$)

$$P = p \cdot t \cdot \sigma_t$$
$$= 65 \times 6 \times 120$$
$$= 46800 \text{ N}$$

**7. Tearing Strength — Double Lap ($P_t$)**

Net section tensile capacity

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (65 - 20) \times 6 \times 120$$
$$= 32400 \text{ N}$$

**8. Shearing Strength — Double Lap ($P_s$)**

Single shear capacity ($n = 2$)

$$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 2 \times \frac{\pi}{4} (20)^2 \times 90$$
$$= 56548 \text{ N}$$

**9. Crushing Strength — Double Lap ($P_c$)**

Bearing capacity ($n = 2$)

$$P_c = n \cdot d \cdot t \cdot \sigma_c$$
$$= 2 \times 20 \times 6 \times 180$$
$$= 43200 \text{ N}$$

**10. Efficiency — Double Riveted Lap Joint ($\eta_2$)**

Ratio of minimum strength to solid strength

$$\eta_2 = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$
$$= \frac{32400}{46800} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta_2 = 69.2\%}$$

**Design Output**

**Single Riveted Lap Efficiency eta1 = 60.0% | Double Riveted Lap Efficiency eta2 = 69.2%**

# SECTION 2: SINGLE RIVETED LAP JOINT — MECHANICAL DIAGRAM

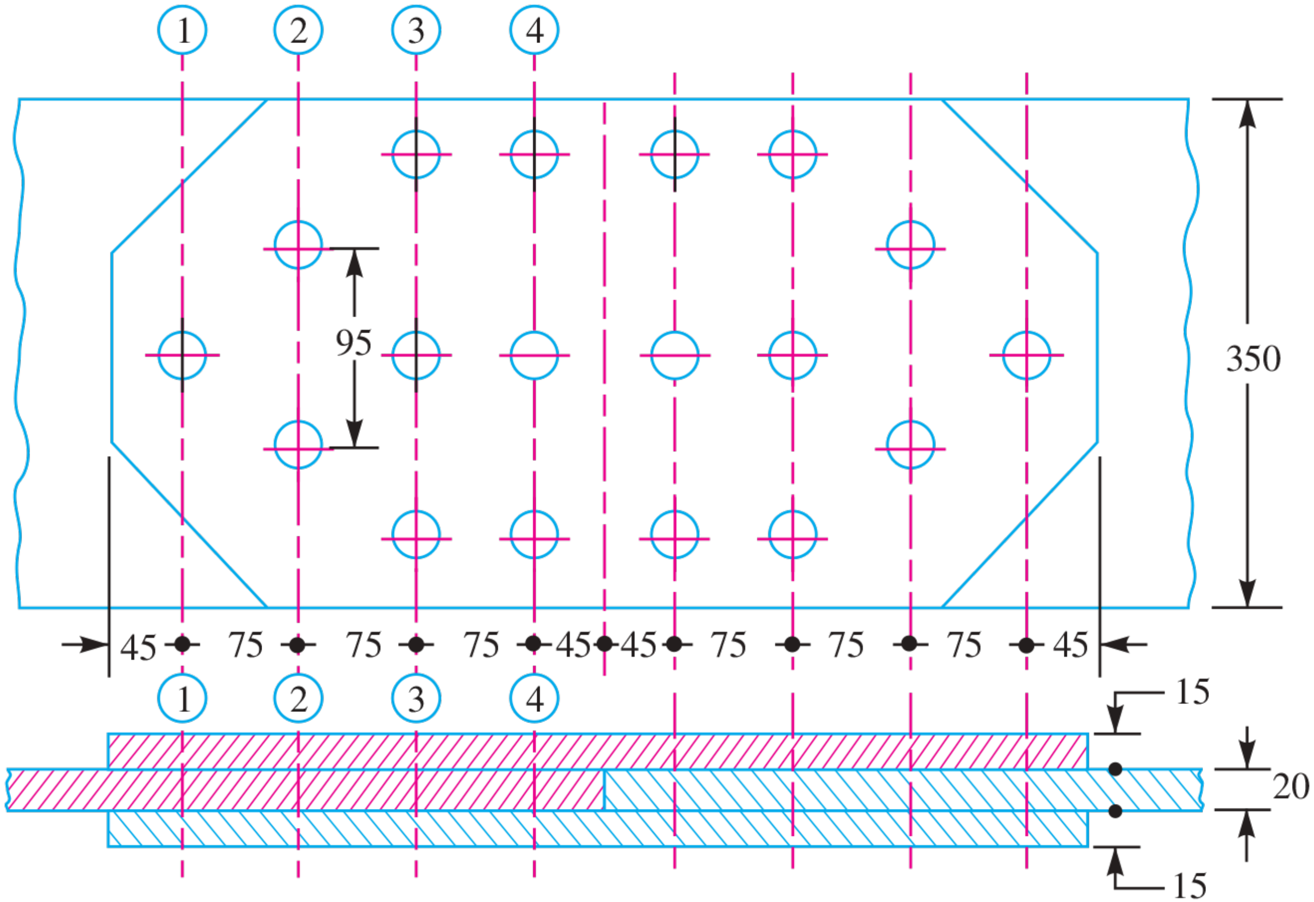


**Fig. 9.21.** All dimensions in mm.

**Fig. 9.21: Single Riveted Lap Joint Scheme**

# SECTION 3: EFFICIENCY OF DOUBLE RIVETED DOUBLE COVER BUTT JOINT

**System Parameters:** Plate Thickness $t = 20$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 100$ mm, Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 100$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 2.0.

**Design Protocol:** Evaluate solid plate strength, plate tearing strength, rivet double shear strength ($n = 2$), and rivet crushing strength to determine joint efficiency.

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**1. Solid Plate Strength ($P$)**

Un-punched solid plate tensile capacity

$$P = p \cdot t \cdot \sigma_t$$
$$= 100 \times 20 \times 120$$
$$= 240000 \text{ N}$$

**2. Tearing Strength of Plate ($P_t$)**

Net section tensile strength per pitch

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (100 - 25) \times 20 \times 120$$
$$= 180000 \text{ N}$$

**3. Shearing Strength of Rivets ($P_s$)**

Double shear capacity of 2 rivets per pitch ($n = 2$, factor = 2)

$$P_s = 2 \cdot \left(2 \cdot \frac{\pi}{4} d^2\right) \cdot \tau$$
$$= 2 \times \left(2 \times \frac{\pi}{4}(25)^2\right) \times 100$$
$$= 196350 \text{ N}$$

**4. Crushing Strength of Rivets ($P_c$)**

Bearing capacity of 2 rivets per pitch ($n = 2$)

$$P_c = n \cdot d \cdot t \cdot \sigma_c$$

$$= 2 \times 25 \times 20 \times 150$$

$$= 150000 \text{ N}$$

**5. Joint Efficiency ($\eta$)**

Ratio of minimum strength to un-punched solid plate strength

$$\eta = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$

$$= \frac{150000}{240000} \times 100\%$$

$$\Rightarrow \boldsymbol{\eta = 62.5}\%$$

**Design Output**

**Double Cover Butt Joint Efficiency eta = 62.5% (Governed by Rivet Crushing)**

# SECTION 3: DOUBLE RIVETED DOUBLE COVER BUTT JOINT — MECHANICAL DIAGRAM

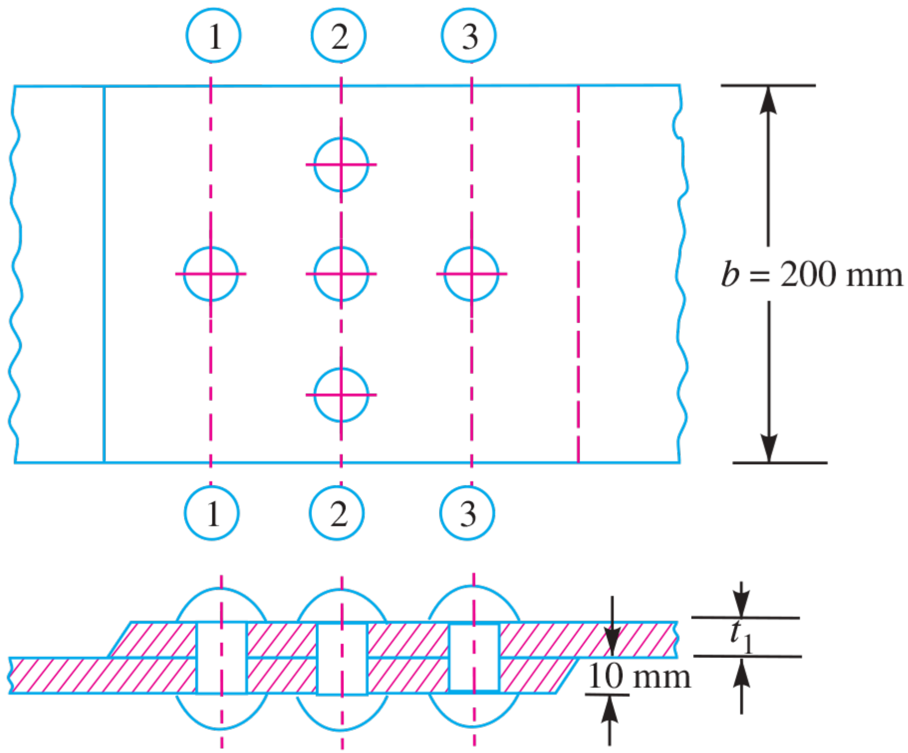


Fig. 9.22

**Fig. 9.22: Double Riveted Double Cover Butt Joint**

# SECTION 4: DOUBLE RIVETED LAP JOINT — ZIG-ZAG RIVETING

**System Parameters:** Plate Thickness $t = 13$ mm, Allowable Stresses: Tensile $\sigma_t = 80$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 120$ MPa, Number of Rivets per pitch $n = 2$.

**Design Protocol:** Determine rivet hole diameter using Unwin's formula, equate tearing strength to shearing strength to solve for pitch $p$, and evaluate joint efficiency.

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**1. Rivet Diameter ($d$)**

Unwin's formula for $t = 13 \text{ mm} \geq 8 \text{ mm}$

$$d = 6\sqrt{t}$$
$$= 6\sqrt{13}$$
$$= 21.63 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 22 \text{ mm}}$$

**2. Selected Rivet Standard Size**

Adopted standard nominal rivet diameter

**Adopted Rivet Diameter d = 22 mm**

**3. Rivet Pitch ($p$)**

Equating tearing strength $P_t$ to shearing strength $P_s$ ($n = 2$)

$$(p - d)t \cdot \sigma_t = n \cdot \frac{\pi}{4}d^2 \cdot \tau$$

$$(p - 22) \times 13 \times 80 = 2 \times \frac{\pi}{4}(22)^2 \times 60$$

$$1040(p - 22) = 45616$$

$$p - 22 = \frac{45616}{1040}$$

$$p - 22 = 43.86$$

$$p = 65.86 \text{ mm}$$

$$\Rightarrow \boldsymbol{p = 66 \text{ mm}}$$

**4. Adopted Pitch Dimension**

Selected longitudinal pitch length

**Adopted Pitch p = 66 mm**

**5. Joint Efficiency ($\eta$)**

Efficiency based on shear strength vs solid plate strength

$$\eta = \frac{P_s}{p \cdot t \cdot \sigma_t} \times 100\%$$

$$= \frac{45616}{66 \times 13 \times 80} \times 100\%$$

$$= \frac{45616}{68640} \times 100\%$$

$$\Rightarrow \boldsymbol{\eta = 66.5}\%$$

**Design Output**

**Zig-zag Lap Joint: Rivet Diameter d = 22 mm | Pitch p = 66 mm | Efficiency eta = 66.5%**

# SECTION 4: DOUBLE RIVETED LAP JOINT (ZIG-ZAG) — MECHANICAL DIAGRAM

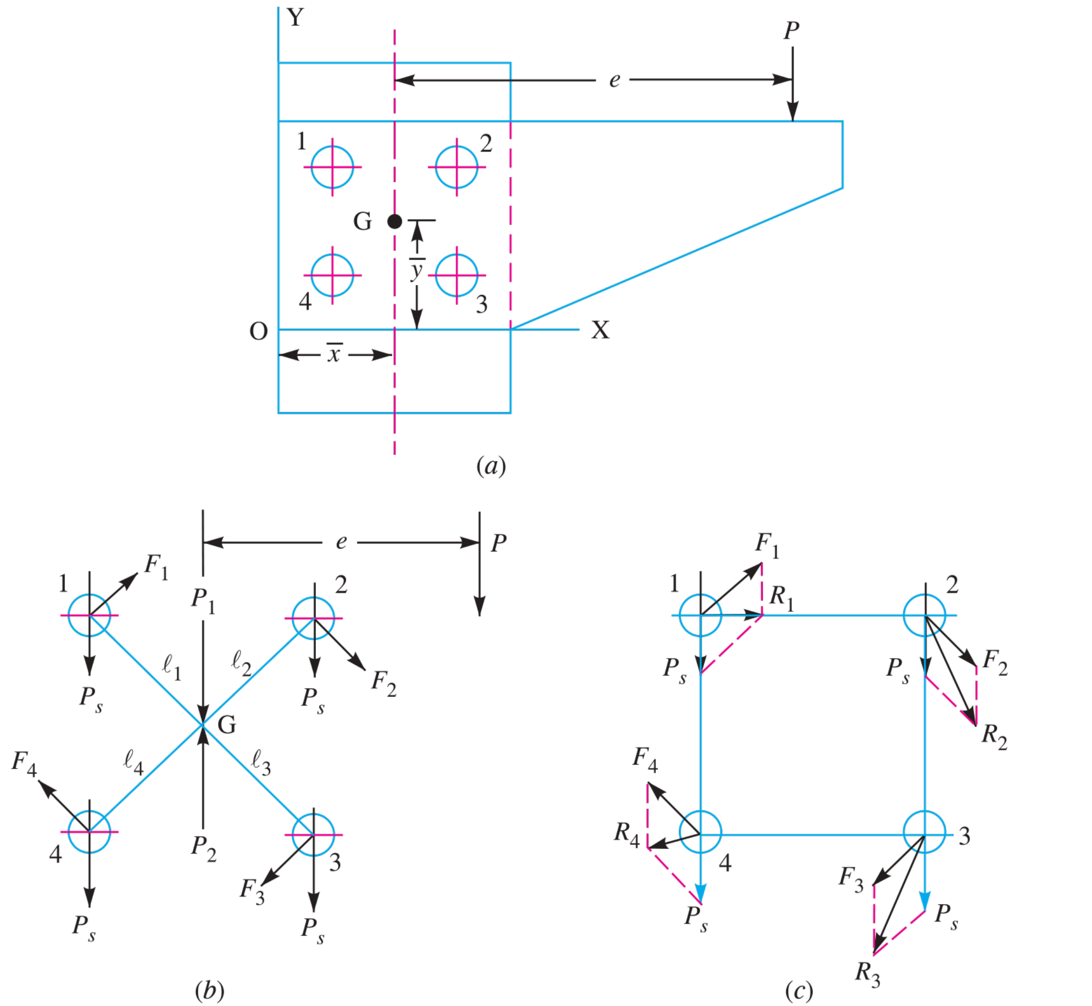


**Fig. 9.23.** Eccentric loaded riveted joint.

**Fig. 9.23: Double Riveted Lap Joint with Zig-Zag Riveting**

# SECTION 5: TRIPLE RIVETED LAP JOINT — ZIG-ZAG PATTERN

**System Parameters:** Plate Thickness $t = 7$ mm, Allowable Stresses: Tensile $\sigma_t = 90$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 120$ MPa, Number of Rivets per pitch $n = 3$.

**Design Protocol:** For thin plates ($t < 8$ mm), equate shearing strength to crushing strength to determine rivet diameter $d$; equate tearing strength to shearing strength to find pitch $p$; compute transverse row pitch $p_b$.

---

**1. Rivet Diameter ($d$)**

Equating shearing strength $P_s$ to crushing strength $P_c$ ($t < 8$ mm)

$$n \cdot \frac{\pi}{4} d^2 \cdot \tau = n \cdot d \cdot t \cdot \sigma_c$$

$$3 \times \frac{\pi}{4} d^2 \times 60 = 3 \times d \times 7 \times 120$$

$$141.37d = 2520$$

$$d = \frac{2520}{141.37}$$

$$d = 17.82 \text{ mm}$$

$$\Rightarrow \boldsymbol{d = 19 \text{ mm}}$$

**2. Selected Rivet Standard Size**

Adopted standard nominal rivet diameter

**Adopted Rivet Diameter d = 19 mm**

**3. Rivet Pitch ($p$)**

Equating tearing strength $P_t$ to shearing strength $P_s$ ($n = 3$)

$$(p-d)t \cdot \sigma_t = n \cdot \frac{\pi}{4}d^2 \cdot \tau$$

$$(p-19) \times 7 \times 90 = 3 \times \frac{\pi}{4}(19)^2 \times 60$$

$$630(p-19) = 51035$$

$$p - 19 = \frac{51035}{630}$$

$$p - 19 = 81.01$$

$$p = 100.01 \text{ mm}$$

$$\Rightarrow \boldsymbol{p = 100 \text{ mm}}$$

**4. Adopted Pitch Dimension**

Selected longitudinal pitch length

**Adopted Pitch p = 100 mm**

**5. Transverse Row Pitch ($p_b$)**

Empirical relation for zig-zag riveting

$$p_b = 0.33p + 0.67d$$

$$= 0.33(100) + 0.67(19)$$

$$= 33 + 12.73$$

$$= 45.73 \text{ mm}$$

$$\Rightarrow \boldsymbol{p_b = 46 \text{ mm}}$$

**Design Output**

**Triple Lap Joint: d = 19 mm | Pitch p = 100 mm | Row Pitch pb = 46 mm**

# SECTION 5: TRIPLE RIVETED LAP JOINT — MECHANICAL DIAGRAM

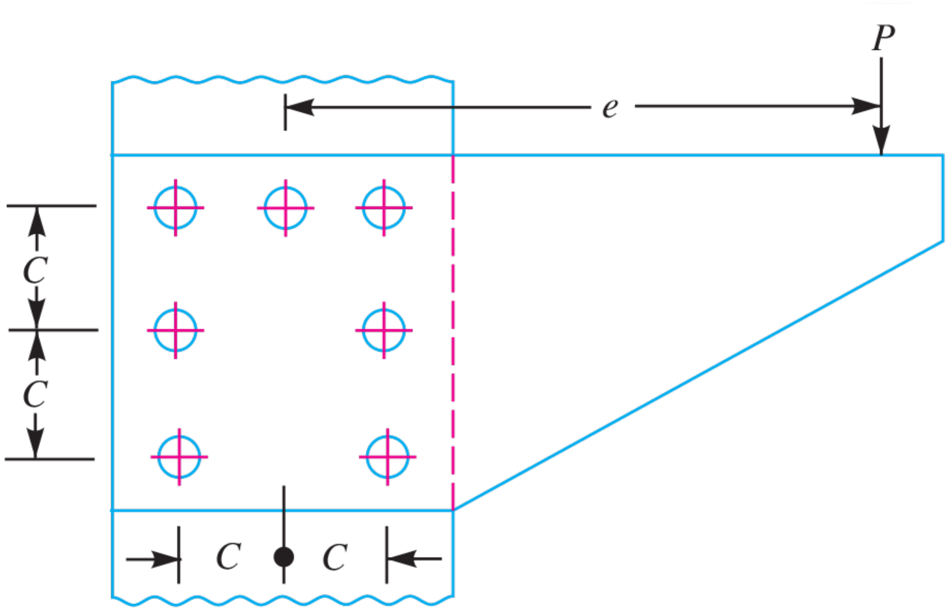


Fig. 9.24

**Fig. 9.24: Triple Riveted Lap Joint with Zig-Zag Pattern**

# SECTION 6: SINGLE RIVETED DOUBLE STRAP BUTT JOINT

**System Parameters:** Main Plate Thickness $t = 10$ mm, Allowable Tensile Stress $\sigma_t = 80$ MPa, Allowable Shear Stress $\tau = 60$ MPa, Double Shear Factor = 1.875.

**Design Protocol:** Determine rivet diameter $d$ using Unwin's formula, calculate double shear capacity $P_s$ ($n = 1$), solve for pitch $p$, compute cover strap thickness $t_1$, and evaluate efficiency.

---

**1. Rivet Diameter ($d$)**

Unwin's formula for $t = 10$ mm

$$d = 6\sqrt{t}$$
$$= 6\sqrt{10}$$
$$= 18.97 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 19 \text{ mm}}$$

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**2. Double Shear Strength ($P_s$)**

IS code double shear strength for single rivet ($n = 1$)

$$P_s = 1.875 \cdot n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1.875 \times 1 \times \frac{\pi}{4}(19)^2 \times 60$$
$$= 31895 \text{ N}$$

**3. Rivet Pitch ($p$)**

Equating tearing strength $P_t$ to double shear strength $P_s$

$$(p-d)t\cdot\sigma_t = P_s$$

$$(p-19)\times 10\times 80 = 31895$$

$$800(p-19) = 31895$$

$$p-19 = \frac{31895}{800}$$

$$p-19 = 39.87$$

$$p = 58.87 \text{ mm}$$

$$\Rightarrow \boldsymbol{p = 60 \text{ mm}}$$

**4. Cover Strap Thickness ($t_1$)**

Standard proportion for double cover butt joint

$$t_1 = 0.625t$$

$$= 0.625\times 10$$

$$= 6.25 \text{ mm}$$

$$\Rightarrow \boldsymbol{t_1 = 7 \text{ mm}}$$

**5. Joint Efficiency ($\eta$)**

Ratio of double shear strength to solid plate strength

$$\eta = \frac{P_s}{p\cdot t\cdot\sigma_t}\times 100\%$$

$$= \frac{31895}{60\times 10\times 80}\times 100\%$$

$$= \frac{31895}{48000}\times 100\%$$

$$\Rightarrow \boldsymbol{\eta = 66}.4\%$$

| | |
|---|---|
| **Design Output** | **Double Strap Butt Joint: d = 19 mm \| Pitch p = 60 mm \| Strap t1 = 7 mm \| Efficiency eta = 66.4%** |

# SECTION 6: SINGLE RIVETED DOUBLE STRAP BUTT JOINT — MECHANICAL DIAGRAM

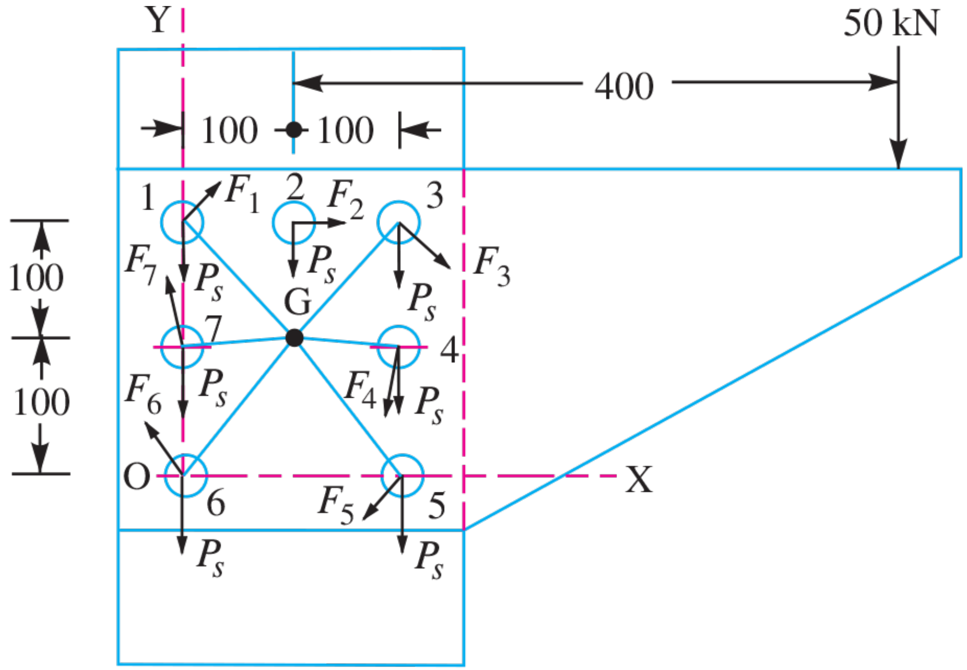


Fig. 9.25

Fig. 9.25: Single Riveted Double Strap Butt Joint

# SECTION 7: BOILER LONGITUDINAL DOUBLE RIVETED BUTT JOINT

**System Parameters:** Boiler Shell Diameter $D = 1.5\ \text{m} = 1500\ \text{mm}$, Internal Steam Pressure $p_i = 0.95\text{N/mm}^2$, Target Longitudinal Efficiency $\eta_l = 75\%$, Allowable Tensile Stress $\sigma_t = 90\ \text{MPa}$, Allowable Crushing Stress $\sigma_c = 140\ \text{MPa}$, Allowable Shear Stress $\tau = 56\ \text{MPa}$, Corrosion Allowance = 1.5 mm.

**Design Protocol:** Calculate shell thickness $t$, size rivet diameter $d$, solve pitch $p$ from double shear strength ($n = 2$), check against maximum pitch $p_{\text{max}}$, and summarize design parameters.

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**1. Shell Plate Thickness ($t$)**

Thin cylinder formula with corrosion allowance

$$t = \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} + 1.5$$

$$= \frac{0.95 \times 1500}{2 \times 90 \times 0.75} + 1.5$$

$$= \frac{1425}{135} + 1.5$$

$$= 10.55 + 1.5$$

$$= 12.05\ \text{mm}$$

$$\Rightarrow \boldsymbol{t = 13\ \text{mm}}$$

**2. Selected Shell Thickness**

Adopted standard plate thickness

**Adopted Shell Thickness t = 13 mm**

**3. Rivet Diameter ($d$)**

Unwin's formula for $t = 13$ mm

$$d = 6\sqrt{t}$$
$$= 6\sqrt{13}$$
$$= 21.63 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 22 \text{ mm}}$$

**4. Selected Rivet Diameter**

Adopted standard nominal rivet size

**Adopted Rivet Diameter d = 22 mm**

**5. Double Shear Strength ($P_s$)**

IS double shear capacity for $n = 2$ rivets per pitch

$$P_s = 1.875 \cdot n \cdot \frac{\pi}{4} d^2 \cdot \tau$$
$$= 1.875 \times 2 \times \frac{\pi}{4}(22)^2 \times 56$$
$$= 79828 \text{ N}$$

**6. Rivet Pitch ($p$)**

Calculated pitch from $P_t = P_s$ and maximum pitch check ($p_{\text{max}}$)

$$(p - d)t \cdot \sigma_t = P_s$$

$$\begin{aligned} (p-22) \times 13 \times 90 &= 79828 \\ 1170(p-22) &= 79828 \\ p - 22 &= 68.23 \\ p &= 90.23 \text{ mm} \\ p_{\text{max}} &= 3.5t + 41.28 \\ &= 3.5(13) + 41.28 \\ &= 86.78 \text{ mm} \\ &\Rightarrow \boldsymbol{p = 86 \text{ mm}} \end{aligned}$$

**Design Output**

**Boiler Longitudinal Seam: Shell Thickness t = 13 mm | Rivet Diameter d = 22 mm | Pitch p = 86 mm**

# SECTION 7: BOILER LONGITUDINAL DOUBLE RIVETED BUTT JOINT — MECHANICAL DIAGRAM

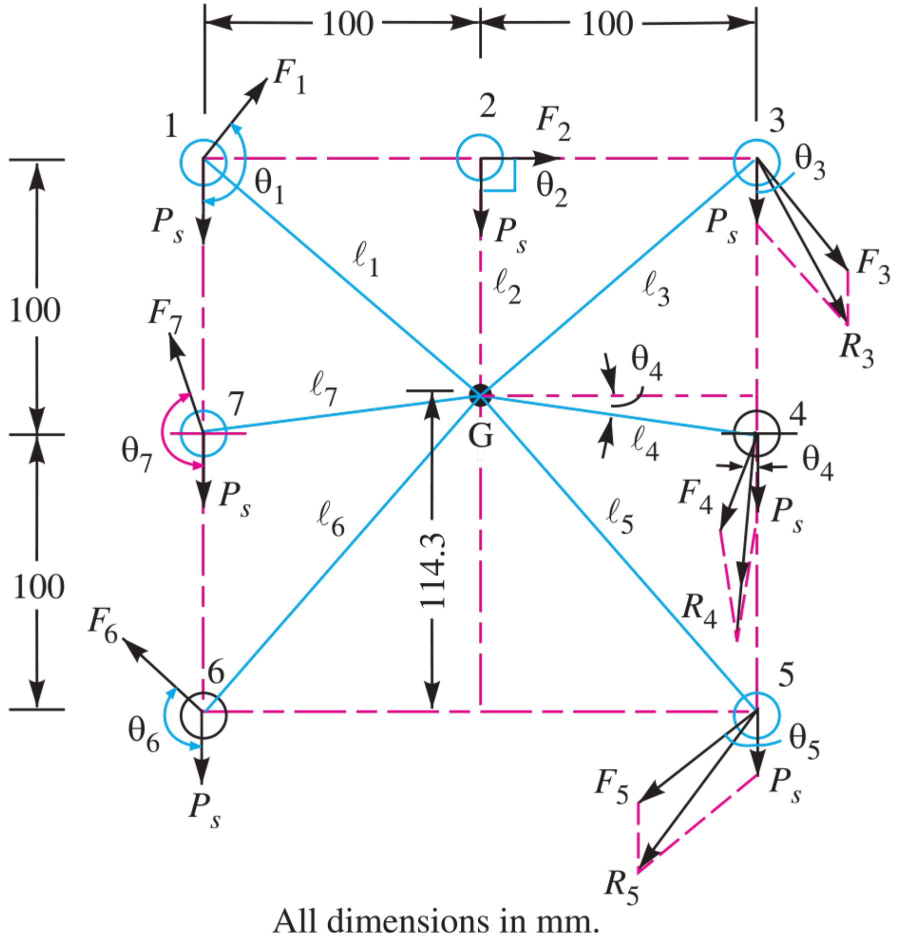


**Fig. 9.26**

**Fig. 9.26: Boiler Longitudinal Double Riveted Double Strap Butt Joint**

# SECTION 8: PRESSURE VESSEL DOUBLE STRAP BUTT JOINT

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**System Parameters:** Vessel Diameter $D = 1\ \mathrm{m} = 1000\ \mathrm{mm}$, Pressure $p_i = 2.75\mathrm{N/mm}^2$, Target Efficiency $\eta_l = 79\%$, Allowable Tensile Stress $\sigma_t = 88\ \mathrm{MPa}$, Allowable Shear Stress $\tau = 64\ \mathrm{MPa}$, Double Shear Factor = 1.8.

**Design Protocol:** Size shell thickness $t$, solve rivet diameter $d$, determine outer and inner row pitches ($p_{\mathrm{outer}}, p_{\mathrm{inner}}$), and verify efficiency.

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**1. Plate Thickness ($t$)**

Thin cylinder formula

$$t = \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} = \frac{2.75 \times 1000}{2 \times 88 \times 0.79} = \frac{2750}{139.04} = 19.78\ \mathrm{mm} \Rightarrow \boldsymbol{t = 20\ \mathrm{mm}}$$

---

**2. Rivet Diameter ($d$)**

Unwin's formula for $t = 20\ \mathrm{mm}$

$$d = 6\sqrt{t} = 6\sqrt{20} = 26.83\ \mathrm{mm} \Rightarrow \boldsymbol{d = 28.5\ \mathrm{mm}}$$

### 3. Outer Pitch ($p$) & Efficiency ($\eta$)

Selected pitch dimensions and verified efficiency

$$p_{\text{outer}} = 150 \text{ mm}$$

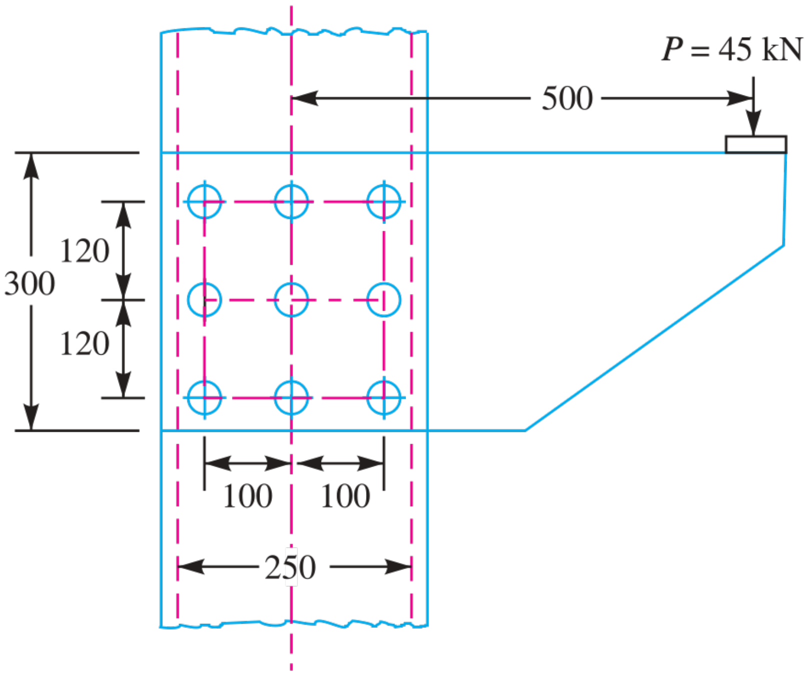
$$p_{\text{inner}} = 75 \text{ mm}$$

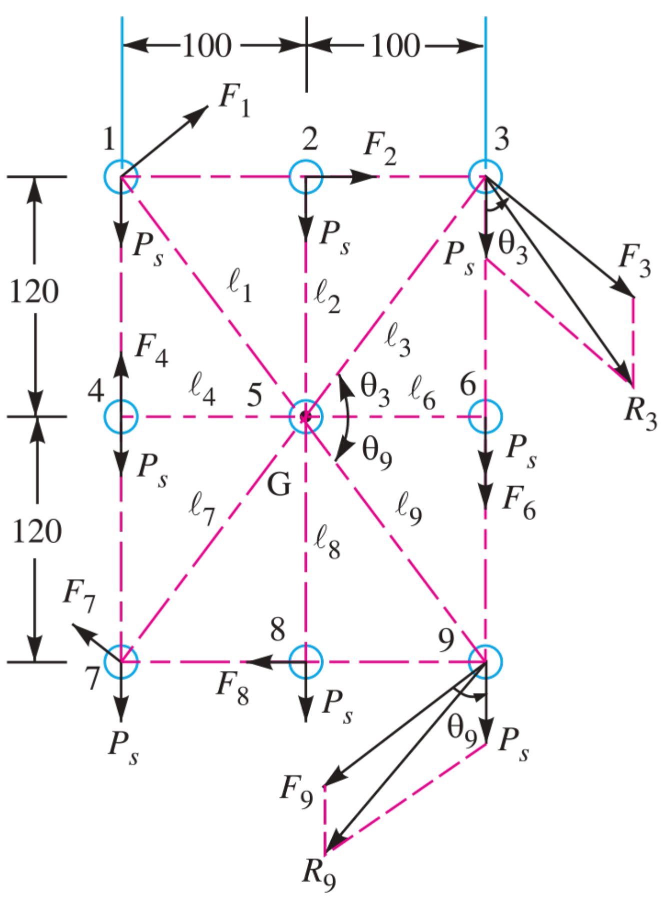
$$\begin{aligned} P_t &= (p_{\text{outer}} - d)t \cdot \sigma_t \\ &= (150 - 28.5) \times 20 \times 88 \\ &= 213840 \text{ N} \\ P &= 150 \times 20 \times 88 \\ &= 264000 \text{ N} \\ \eta &= \frac{213840}{264000} \times 100\% \\ &\Rightarrow \boldsymbol{\eta = 81\%} \end{aligned}$$

**Design Output**

**Pressure Vessel Joint: Thickness t = 20 mm | Rivet Diameter d = 28.5 mm | Outer Pitch p = 150 mm | Efficiency eta = 81%**

# SECTION 8: PRESSURE VESSEL BUTT JOINT — MECHANICAL DIAGRAM




All dimensions in mm.

**Fig. 9.27** | **Fig. 9.28**

**Fig. 9.27 & 9.28: Pressure Vessel Double Strap Butt Joint**

# SECTION 9: BOILER LONGITUDINAL SEAM DESIGN

**System Parameters:** Boiler Diameter $D = 1.25\ \mathrm{m} = 1250\ \mathrm{mm}$, Steam Pressure $p_i = 2.5\mathrm{N/mm}^2$, Ultimate Tensile Stress $\sigma_{\mathrm{u}} = 420\ \mathrm{MPa}$, Ultimate Crushing Stress $\sigma_{\mathrm{cu}} = 650\ \mathrm{MPa}$, Ultimate Shear Stress $\tau_{\mathrm{u}} = 300\ \mathrm{MPa}$, Target Efficiency $\eta_l = 80\%$, Factor of Safety $\mathrm{FOS} = 5$, Corrosion Allowance = 1.5 mm.

**Design Protocol:** Calculate working allowable stresses, determine plate thickness $t$, and compute rivet diameter $d$.

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**1. Allowable Working Stresses**

Ultimate stress divided by FOS = 5

$$\sigma_t = \frac{\sigma_{\mathrm{u}}}{\mathrm{FOS}} = \frac{420}{5} = 84\ \mathrm{MPa}$$

$$\sigma_c = \frac{\sigma_{\mathrm{cu}}}{\mathrm{FOS}} = \frac{650}{5} = 130\ \mathrm{MPa}$$

$$\tau = \frac{\tau_{\mathrm{u}}}{\mathrm{FOS}} = \frac{300}{5} = 60\ \mathrm{MPa}$$

**2. Plate Thickness ($t$)**

Thin cylinder formula with corrosion allowance

$$t = \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} + 1.5$$

$$= \frac{2.5 \times 1250}{2 \times 84 \times 0.80} + 1.5$$

$$= \frac{3125}{134.4} + 1.5$$

$$= 23.25 + 1.5$$

$$= 24.75 \text{ mm}$$

$$\Rightarrow \boldsymbol{t = 25 \text{ mm}}$$

**3. Rivet Diameter ($d$)**

Unwin's formula for $t = 25$ mm

$$d = 6\sqrt{t}$$

$$= 6\sqrt{25}$$

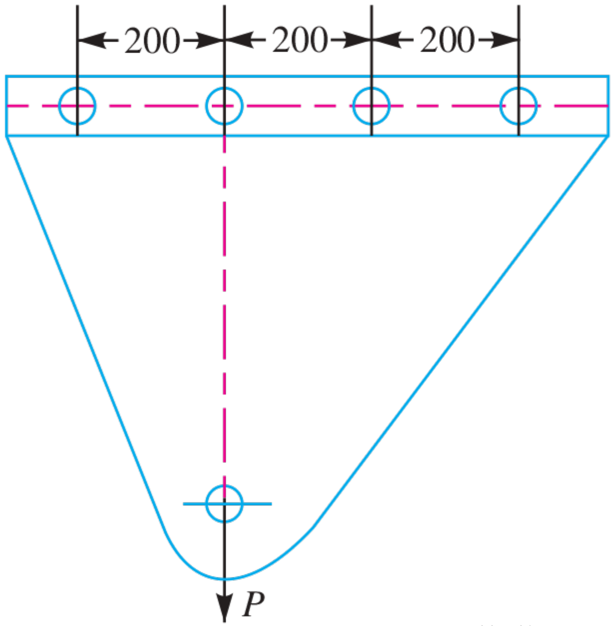
$$= 30 \text{ mm}$$

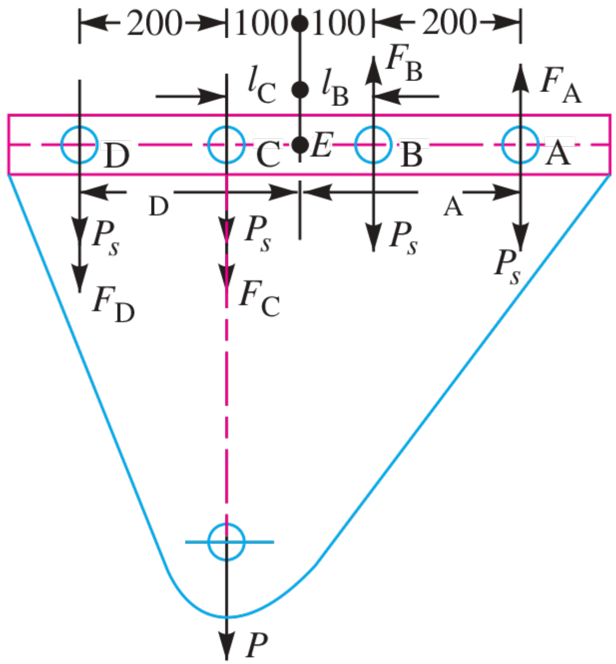
$$\Rightarrow \boldsymbol{d = 30 \text{ mm}}$$

**Design Output**

**Boiler Seam: Allowable Stresses (84, 130, 60 MPa) | Plate Thickness t = 25 mm | Rivet Diameter d = 30 mm**

# SECTION 9: BOILER LONGITUDINAL SEAM — MECHANICAL DIAGRAM




All dimensions in mm.

**Fig. 9.29** **Fig. 9.30**

**Fig. 9.29 & 9.30: Boiler Longitudinal Seam Triple Riveted Butt Joint**

# SECTION 10: BOILER LONGITUDINAL & CIRCUMFERENTIAL JOINTS

**System Parameters:** Boiler Diameter $D = 1.6\ \mathrm{m} = 1600\ \mathrm{mm}$, Pressure $p_i = 2.5\mathrm{N/mm}^2$, Allowable Stresses: Tensile $\sigma_t = 75\ \mathrm{MPa}$, Shear $\tau = 60\ \mathrm{MPa}$, Crushing $\sigma_c = 125\ \mathrm{MPa}$.

**Design Protocol:** Summarize longitudinal joint plate thickness $t$ and rivet diameter $d$; design circumferential joint by calculating total steam thrust $F$ and required rivet count $N_c$.

---

**1. Longitudinal Joint Parameters**
Plate thickness and rivet diameter selection

$$t = 35\ \mathrm{mm}$$
$$d = 35.5\ \mathrm{mm}$$

**2. Total Steam End Thrust ($F$)**
Axial burst force acting on boiler end cap

$$F = \frac{\pi}{4} D^2 \cdot p_i$$
$$= \frac{\pi}{4}(1600)^2 \times 2.5$$
$$= 5.026 \times 10^6\ \mathrm{N}$$

**3. Number of Circumferential Rivets ($N_c$)**

Number of single shear rivets required to resist axial thrust

$$N_c = \frac{F}{\frac{\pi}{4}d^2 \cdot \tau}$$

$$= \frac{5.026 \times 10^6}{\frac{\pi}{4}(35.5)^2 \times 60}$$

$$= \frac{5.026 \times 10^6}{59390}$$

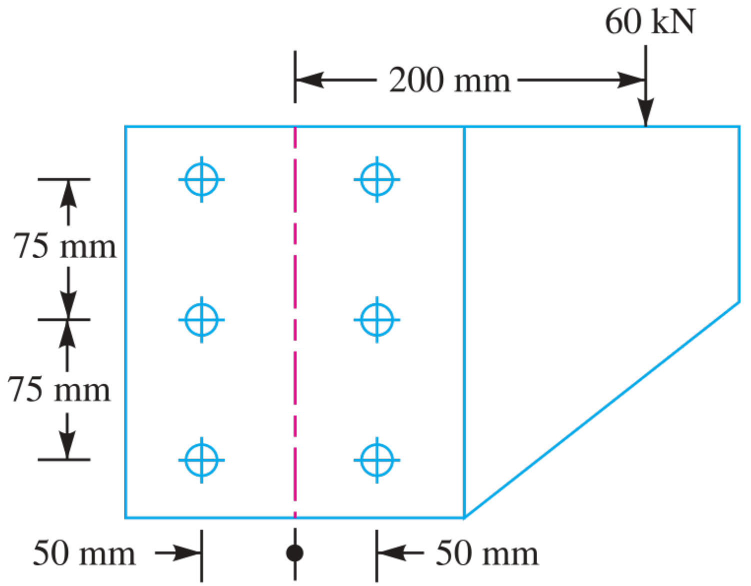
$$= 84.6$$

$$\Rightarrow \boldsymbol{N_c = 85 \text{ rivets}}$$

**Design Output**

**Longitudinal: t = 35 mm, d = 35.5 mm | Circumferential Joint Thrust F = 5.026 MN | Rivet Count Nc = 85 rivets**

# SECTION 10: BOILER CIRCUMFERENTIAL & LONGITUDINAL JOINTS — MECHANICAL DIAGRAM



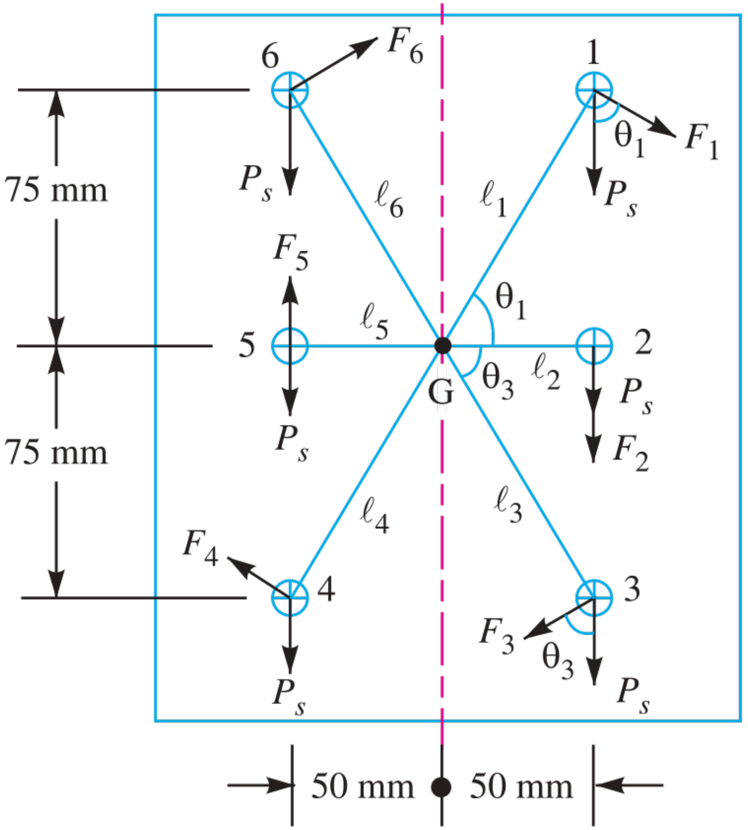




**Fig. 9.31** **Fig. 9.32**

**Fig. 9.31 & 9.32: Boiler Longitudinal and Circumferential Joints**

# SECTION 11: TIE ROD DOUBLE COVER BUTT JOINT

**System Parameters:** Tie Rod Cross-Section $b = 200\ \text{mm}$, $t = 12.5\ \text{mm}$, Allowable Stresses: Tensile $\sigma_t = 80\ \text{MPa}$, Shear $\tau = 65\ \text{MPa}$, Crushing $\sigma_c = 160\ \text{MPa}$, Double Shear Factor = 1.875.

**Design Protocol:** Calculate rivet diameter $d$, evaluate maximum tensile load capacity $P$, compute single rivet double shear strength $P_s$, and determine required rivet count $n$.

---

**1. Rivet Diameter ($d$)**

Unwin's formula for $t = 12.5\ \text{mm}$

$$d = 6\sqrt{t}$$
$$= 6\sqrt{12.5}$$
$$= 21.21\ \text{mm}$$
$$\Rightarrow \boldsymbol{d = 21.5\ \text{mm}}$$

**2. Full Tensile Load ($P$)**

Solid tie-rod tensile capacity

$$P = b \cdot t \cdot \sigma_t$$
$$= 200 \times 12.5 \times 80$$
$$= 200000\ \text{N}$$
$$= 200\ \text{kN}$$

**3. Single Rivet Double Shear Capacity ($P_s$)**

Shear strength per rivet in double cover butt joint

$$P_s = 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$

$$= 1.875 \times \frac{\pi}{4}(21.5)^2 \times 65$$

$$= 44250 \text{ N}$$

$$= 44.25 \text{ kN}$$

**4. Required Rivet Count ($n$)**

Total rivets needed per side of joint

$$n = \frac{P}{P_s}$$

$$= \frac{200}{44.25}$$

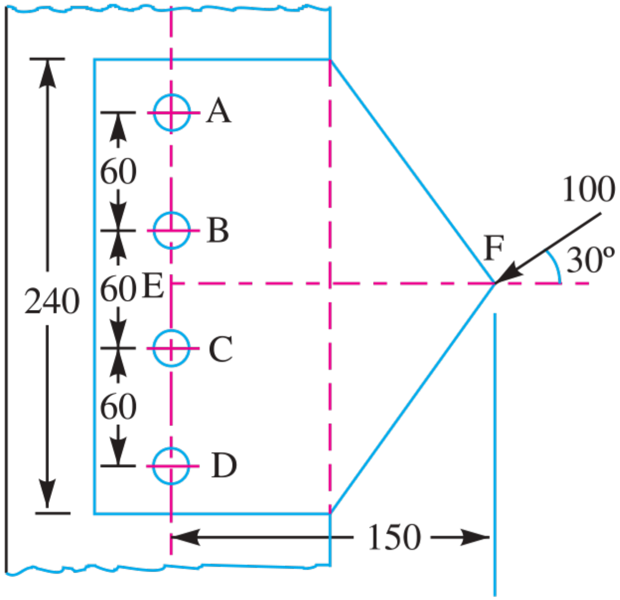
$$= 4.52$$

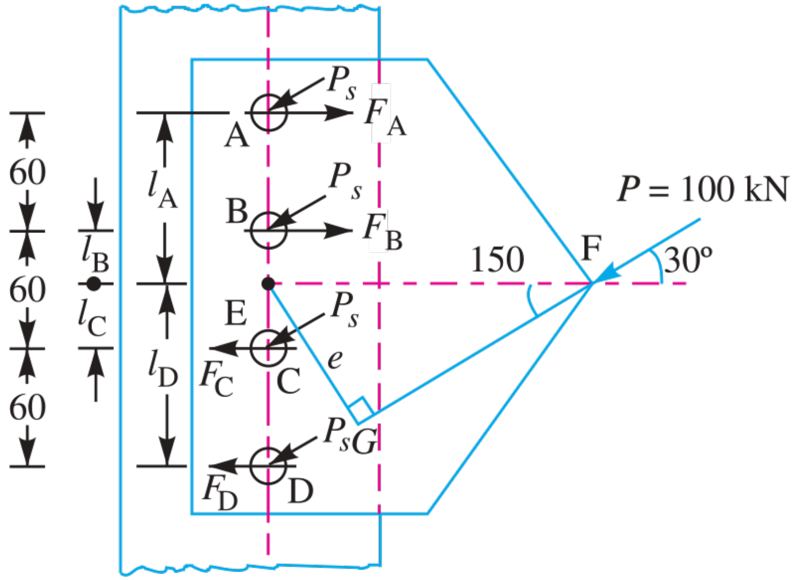
$$\Rightarrow \boldsymbol{n = 5 \text{ rivets}}$$

**Design Output**

**Tie Rod Joint: Rivet Diameter d = 21.5 mm | Tensile Load P = 200 kN | Required Rivets n = 5 rivets**

# SECTION 11: TIE ROD DOUBLE COVER BUTT JOINT — MECHANICAL DIAGRAM




All dimensions in mm.

**Fig. 9.33** **Fig. 9.34**

**Fig. 9.33 & 9.34: Tie Rod Double Cover Butt Joint**

# SECTION 12: BRIDGE TIE-BAR LOZENGE (DIAMOND) JOINT

**System Parameters:** Flat Bar Width $b = 350$ mm, Thickness $t = 20$ mm, Allowable Stresses: Tensile $\sigma_t = 90$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 1.875.

**Design Protocol:** Calculate rivet diameter $d$, evaluate net load capacity $P$ at outer row (1 rivet), determine single rivet double shear strength $P_s$, compute required number of rivets $n$, and arrange in diamond pattern.

---

**1. Rivet Diameter ($d$)**

Unwin's formula for $t = 20$ mm

$$d = 6\sqrt{t}$$
$$= 6\sqrt{20}$$
$$= 26.83 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 27 \text{ mm}}$$

**2. Net Load Capacity ($P$)**

Tearing strength at critical Section 1-1 (1 rivet hole)

$$P = (b - d)t \cdot \sigma_t$$
$$= (350 - 27) \times 20 \times 90$$
$$= 323 \times 1800$$
$$= 581400 \text{ N}$$
$$= 581.4 \text{ kN}$$

**3. Single Rivet Double Shear Capacity ($P_s$)**

Shear resistance per rivet

$$P_s = 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$$

$$= 1.875 \times \frac{\pi}{4}(27)^2 \times 60$$

$$= 64412 \text{ N}$$

$$= 64.4 \text{ kN}$$

**4. Total Rivet Count ($n$)**

Total rivets required for lozenge arrangement

$$n = \frac{P}{P_s}$$

$$= \frac{581.4}{64.4}$$

$$= 9.02$$

$$\Rightarrow \boldsymbol{n = 9 \text{ rivets}}$$

**5. Lozenge Rivet Layout**

Symmetric diamond pattern from outer row to inner row

**Arranged in Diamond Pattern: Row 1 (1 rivet), Row 2 (2 rivets), Row 3 (3 rivets), Row 4 (3 rivets)**

**Design Output**

**Bridge Lozenge Joint: d = 27 mm | Net Capacity P = 581.4 kN | Total Rivets n = 9 (Pattern: 1, 2, 3, 3)**

# SECTION 12: LOZENGE (DIAMOND) JOINT — MECHANICAL DIAGRAM

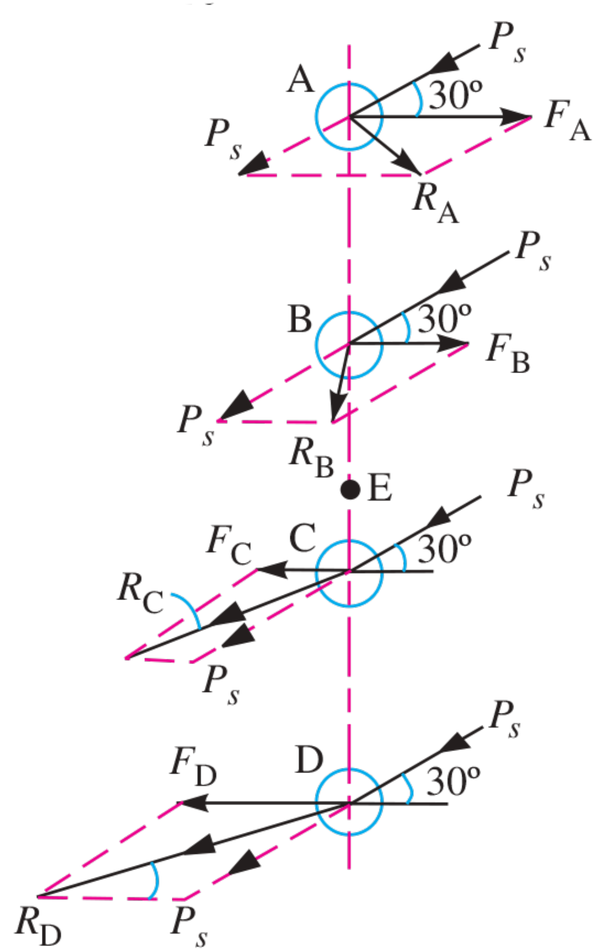



**Fig. 9.35**

**Fig. 9.35: Bridge Tie-Bar Lozenge Joint**

# SECTION 13: FLAT TIE-BAR LAP JOINT (DIAMOND PATTERN)

**System Parameters:** Flat Bar Width $b = 200$ mm, Thickness $t = 10$ mm, Rivet Shank Diameter $d = 24$ mm, Rivet Hole Diameter $d_h = 25.5$ mm, Allowable Stresses: Tensile $\sigma_t = 112$ MPa, Shear $\tau = 84$ MPa, Crushing $\sigma_c = 200$ MPa.

**Design Protocol:** Arrange 4 rivets in diamond pattern (1, 1, 2); evaluate net tearing strength at outer single-rivet row, and compute joint efficiency.

| | |
|---|---|
| **1. Rivet Layout & Pattern**<br>Specified rivet arrangement on flat tie-bar | **n = 4 rivets arranged in Diamond Pattern (Row 1: 1, Row 2: 1, Row 3: 2)** |
| **2. Net Tearing Strength ($P_t$)**<br>Tearing capacity at Section 1-1 with hole diameter $d_h = 25.5$ mm | $P_t = (b - d_h)t \cdot \sigma_t$<br>$= (200 - 25.5) \times 10 \times 112$<br>$= 174.5 \times 1120$<br>$= 195440$ N<br>$= 195.44$ kN |

**3. Solid Strength ($P$) & Efficiency ($\eta$)**

Efficiency based on net tearing capacity

$$P = b \cdot t \cdot \sigma_t$$
$$= 200 \times 10 \times 112$$
$$= 224000 \text{ N}$$
$$\eta = \frac{P_t}{P} \times 100\%$$
$$= \frac{195440}{224000} \times 100\%$$
$$\Rightarrow \boldsymbol{\eta = 87.3\%}$$

**Design Output**

**Flat Lap Joint (Diamond Pattern): Net Tearing Strength Pt = 195.44 kN | Joint Efficiency eta = 87.3%**

# SECTION 13: FLAT TIE-BAR LAP JOINT — MECHANICAL DIAGRAM

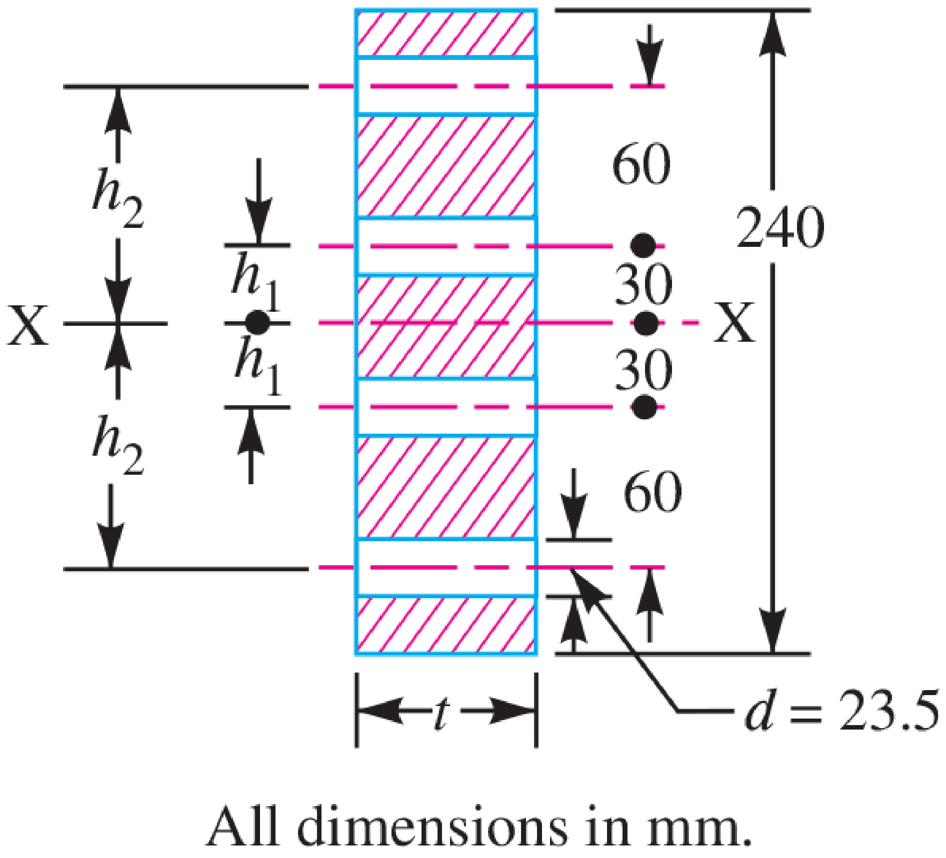



Fig. 9.36

**Fig. 9.36: Flat Tie-Bar Lap Joint (Diamond Pattern)**

# SECTION 14: ECCENTRIC BRACKET WITH VERTICAL RIVET LINE

**System Parameters:** Plate Thickness $t = 25\ \text{mm}$, Eccentric Load $P = 50\ \text{kN} = 50000\ \text{N}$, Eccentricity $e = 400\ \text{mm}$, Rivet Spacing $C = 100\ \text{mm}$, Number of Rivets $n = 4$ in vertical line, Allowable Shear Stress $\tau = 65\ \text{MPa}$, Allowable Crushing Stress $\sigma_c = 120\ \text{MPa}$.

**Design Protocol:** Calculate direct shear force $F_{\text{s1}}$ per rivet; compute secondary shear force $F_{\text{s2}}$ due to moment $Pe$; find max resultant shear force $F_R$, and solve for rivet diameter $d$.

### 1. Direct Shear Force ($F_{\text{s1}}$)

Uniform load distribution across 4 rivets

$$F_{\text{s1}} = \frac{P}{n}$$
$$= \frac{50000}{4}$$
$$= 12500\ \text{N}$$
$$= 12.5\ \text{kN}$$

**2. Secondary Shear Force ($F_{s2}$)**

Torsional shear force on outermost rivet ($r_{\max} = 150$ mm)

$$\sum r_i^2 = 2(150^2 + 50^2)$$
$$= 2(22500 + 2500)$$
$$= 50000\text{mm}^2$$
$$F_{s2} = \frac{P \cdot e \cdot r_{\max}}{\sum} r_i^2$$
$$= \frac{50000 \times 400 \times 150}{50000}$$
$$= 60000 \text{ N}$$
$$= 60 \text{ kN}$$

**3. Maximum Resultant Shear Force ($F_R$)**

Direct plus secondary shear force (co-linear theta = 0 deg)

$$F_R = F_{s1} + F_{s2}$$
$$= 12.5 + 60$$
$$= 72.5 \text{ kN}$$
$$= 72500 \text{ N}$$

### 4. Rivet Diameter ($d$)

Equating $F_R$ to rivet single shear strength

$$F_R = \frac{\pi}{4} d^2 \cdot \tau$$

$$72500 = \frac{\pi}{4} d^2 (65)$$

$$d^2 = \frac{4 \times 72500}{\pi \times 65}$$

$$= \frac{290000}{204.2}$$

$$= 1420.2$$

$$d = 37.68 \text{ mm}$$

$$\Rightarrow \boldsymbol{d = 38 \text{ mm}}$$

**Design Output**

**Eccentric Bracket: Direct Shear Fs1 = 12.5 kN | Secondary Shear Fs2 = 60 kN | Resultant FR = 72.5 kN | d = 38 mm**

# SECTION 15: ECCENTRICALLY LOADED BRACKET JOINT

**System Parameters:** Eccentric Load $P = 45$ kN, Allowable Shear Stress $\tau \leq 40$ MPa.

**Design Protocol:** Evaluate direct and secondary shear forces to determine maximum resultant shear load $F_R = 15.2$ kN; size rivet diameter $d$.

---

**1. Maximum Resultant Shear Force ($F_R$)**

Critical rivet resultant load from direct and secondary shear

$$F_R = 15.2 \text{ kN}$$
$$= 15200 \text{ N}$$

---

**2. Rivet Diameter ($d$)**

Single shear area calculation for allowable stress $\tau = 40$ MPa

$$d = \sqrt{\frac{4F_R}{\pi\tau}}$$
$$= \sqrt{\frac{4 \times 15200}{\pi \times 40}}$$
$$= \sqrt{\frac{60800}{125.66}}$$
$$= \sqrt{483.8}$$
$$= 22.00 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 22} \textbf{ mm}$$

## Design Output

## Eccentric Bracket (45 kN Load): Critical Resultant FR = 15.2 kN | Rivet Diameter d = 22 mm

# SECTION 16: MAXIMUM SAFE ECCENTRIC LOAD CAPACITY

**System Parameters:** Number of Rivets $n = 4$, Rivet Diameter $d = 20\ \mathrm{mm}$, Working Shear Stress $\tau = 100\ \mathrm{MPa}$.

**Design Protocol:** Determine maximum single rivet shear load capacity $F_{\mathrm{R,max}}$; express critical resultant force as a function of external load $P$ ($F_R = 0.966P$); solve maximum safe load $P$.

---

**1. Single Rivet Shear Capacity ($F_{\mathrm{R,max}}$)**

Maximum permissible shear force per rivet

$$F_{\mathrm{R,max}} = \frac{\pi}{4} d^2 \cdot \tau$$

$$= \frac{\pi}{4}(20)^2 \times 100$$

$$= 31416\ \mathrm{N}$$

$$= 31.416\ \mathrm{kN}$$

---

**2. Resultant Load Relation**

Critical rivet resultant load expressed in terms of load $P$

$$F_R = 0.966P$$

**3. Maximum Safe Load ($P$)**

Equating $F_R$ to single rivet shear capacity

$$P = \frac{F_{\mathrm{R,max}}}{0.966} = \frac{31.416}{0.966} = 32.52\ \mathrm{kN}$$

$$\Rightarrow \boldsymbol{P = 32.5\ \mathrm{kN}}$$

**Design Output**

**Eccentric Load Capacity: Single Rivet Capacity**

**FR,max = 31.416 kN | Maximum Safe Load P = 32.5 kN**

# SECTION 17: ECCENTRIC COLUMN BRACKET JOINT

---

**System Parameters:** Number of Rivets $n = 6$, Eccentric Load $P = 60\ \text{kN} = 60000\ \text{N}$, Eccentricity $e = 200\ \text{mm}$, Allowable Shear Stress $\tau \leq 150\ \text{MPa}$.

**Design Protocol:** Calculate primary direct shear $F_{\text{s1}}$ and secondary shear $F_{\text{s2}}$; vectorially combine to get critical resultant shear force $F_R$; solve rivet diameter $d$.

---

### 1. Primary & Secondary Shear Forces

Direct load per rivet and torsional secondary load

$$F_{\text{s1}} = \frac{P}{n} = \frac{60}{6} = 10\ \text{kN}$$

$$F_{\text{s2}} = \frac{P \cdot e \cdot r_{\text{max}}}{\sum} r_i^2 = 41.5\ \text{kN}$$

---

### 2. Maximum Resultant Shear Force ($F_R$)

Vector addition of primary and secondary shear

$$F_R = 51.5\ \text{kN} = 51500\ \text{N}$$

---

### 3. Rivet Diameter ($d$)

Equating $F_R$ to single shear capacity

$$d = \sqrt{\frac{4F_R}{\pi\tau}}$$

$$= \sqrt{\frac{4 \times 51500}{\pi \times 150}}$$

$$= \sqrt{\frac{206000}{471.24}}$$

$$= \sqrt{437.1}$$

$$= 20.91 \text{ mm}$$

$$\Rightarrow \boldsymbol{d = 21 \text{ mm}}$$

**Design Output**

**6-Rivet Column Bracket: Fs1 = 10 kN | Fs2 = 41.5 kN | Resultant FR = 51.5 kN | Rivet Diameter d = 21 mm**

# SECTION 18: ECCENTRICALLY LOADED INCLINED BRACKET JOINT

**System Parameters:** Number of Rivets $n = 4$ ($A, B, C, D$, vertical line, pitch 60 mm), Inclined Load $P = 100\ \text{kN} = 100000\ \text{N}$ at $30^{\text{deg}}$ to horizontal, Eccentricity $e = 150\ \text{mm}$, Allowable Shear Stress $\tau = 80\ \text{MPa}$, Allowable Bending Stress $\sigma_b = 125\ \text{MPa}$, Plate Width $b = 240\ \text{mm}$.

**Design Protocol:** Resolve load into vertical and horizontal components ($P_V, P_H$); calculate max resultant shear force $F_R$ on critical rivet $D$ to size diameter $d$; determine bending moment $M$ on bracket plate to calculate plate thickness $t$.

---

**1. Vertical & Horizontal Load Components**

Resolving inclined 100 kN load at 30 degrees

$$\begin{aligned} P_V &= P\sin\left(30^{\text{deg}}\right) \\ &= 100\sin\left(30^{\text{deg}}\right) \\ &= 50\ \text{kN} \\ P_H &= P\cos\left(30^{\text{deg}}\right) \\ &= 100\cos\left(30^{\text{deg}}\right) \\ &= 86.6\ \text{kN} \end{aligned}$$

## 2. Critical Resultant Shear ($F_R$) & Rivet Diameter ($d$)

Resultant force on rivet D and single shear sizing

$$F_R = 42.4 \text{ kN}$$
$$= 42400 \text{ N}$$
$$d = \sqrt{\frac{4F_R}{\pi\tau}}$$
$$= \sqrt{\frac{4 \times 42400}{\pi \times 80}}$$
$$= \sqrt{\frac{169600}{251.33}}$$
$$= \sqrt{674.8}$$
$$= 25.98 \text{ mm}$$
$$\Rightarrow \boldsymbol{d = 26 \text{ mm}}$$

### 3. Bending Moment ($M$) & Bracket Thickness ($t$)

Bending stress analysis of bracket plate

$$
\begin{aligned}
M &= P_H \cdot e \\
&= 86.6 \times 10^3 \times 150 \\
&= 1.30 \times 10^7 \ \mathrm{N \cdot mm} \\
\sigma_b &= \frac{6M}{t \cdot b^2} \\
125 &= \frac{6 \times 1.30 \times 10^7}{t \times (240)^2} \\
&= \frac{7.80 \times 10^7}{57600t} \\
&= \frac{1354.17}{t} \\
t &= \frac{1354.17}{125} \\
&= 10.83 \ \mathrm{mm} \\
&\Rightarrow \boldsymbol{t = 14 \ \mathrm{mm}}
\end{aligned}
$$

**Design Output**

**Inclined Bracket Joint: PV = 50 kN, PH = 86.6 kN | Critical FR = 42.4 kN | d = 26 mm | Plate Thickness t = 14 mm**